

AI in research and teaching

A critical introduction

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In two hours you should know...

1. **How to spot it** (and accepting that you can't)
2. **How to educate and research** in an AI world
3. **How to judge it** (and reduce risks)

It’s been almost four years

94%

“94% say they use generative AI to help with assessed work. Students directly *including* AI-generated text in assessed work has risen to 12%, up from 8% in 2025 and 3% in 2024.”
Higher Education Policy Institute (HEPI) n1054

(How) are you using it?

“Three key dynamics”

1. “A paradox between **widespread use** and persistent concern;
2. “**Conflicting expectations** among individual scholars, institutions, and journals;
3. “The influence of **cultural and linguistic contexts** on adoption patterns.”

Ho, J. C. et al. (2026) ‘Guidance over guidelines? Unpacking the uses and concerns of generative AI in communication science’, Information, Communication & Society, pp. 1–26. doi: 10.1080/1369118X.2026.2736810.

What is GenAI

What is GenAI (not)?

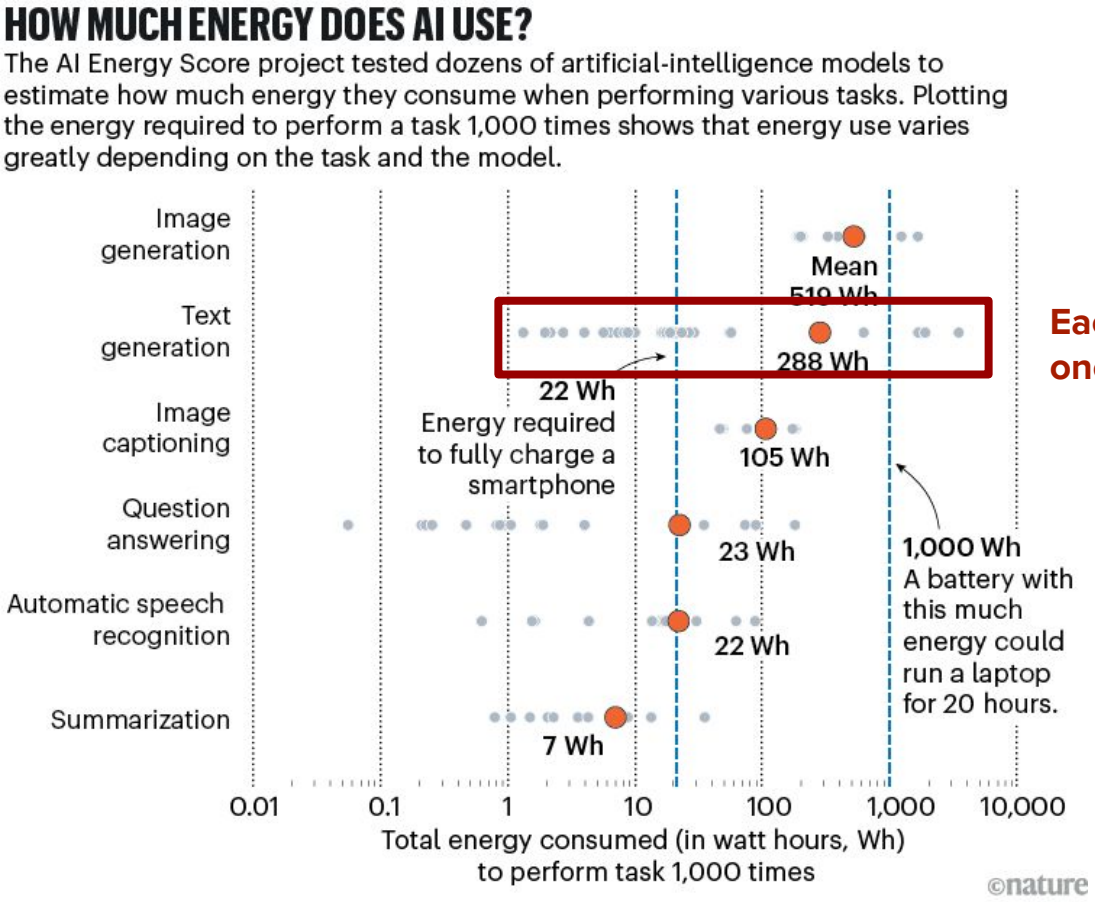
21. PhotoRoom	31. PIXAI	41. MaxAI.me
22. YODAYO	32. ideogram	42. Craiyon
23. Clipchamp	33. invideo AI	43. OpusClip
24. runway	34. Replicate	44. BLACKBOX AI
25. YOU	35. Playground	45. CHATPDF
26. DeepAI	36. Suno	46. PIXELCUT
27. Eightify	37. Chub.ai	47. Vectorizer.AI
28. candy.ai	38. Speechify	48. DREAMGF
29. NightCafé	39. phind	49. Photomyne
30. VocalRemover	40. NovelAI	50. Otter.ai

It's not ChatGPT

Model	Features: Context Window	Creator	Intelligence: Artificial Analysis Intelligence Index	Price: Cost per Task USD	Speed: Median Tokens/s	Latency: Latency First Chunk (s)
Claude Opus 5.5 (max with fallback)	1M	Anthropic	58	$5.98	95	673.65
Claude Sonnet 5.5 (max with fallback)	1M	Anthropic	56	$5.46	136	438.45
Claude Opus 5.5 (xhigh with fallback)	1M	Anthropic	56	$3.46	83	154.25
Muse Wonder Full (max)	1M	Meta				
Claude Opus 5.5 (high with fallback)	1M	Anthropi				
Claude Fable 5.1 (max with fallback)	1M	Anthropi				
Claude Fable 5.1 (xhigh with fallback)	1M	Anthropi				
GPT-6 Astra (max)	1M	OpenAI				
Gemini 4 Argon (high)	1M	Google				

Model	Released	Status now
GPT-3.5	Nov 2022	Retired — replaced by the GPT-5 family
GPT-4	Mar 2023	Retired
GPT-4o	May 2024	Retired from ChatGPT
GPT-4.1	2025	Retired from ChatGPT
o-series (o1/o3/o4-mini)	2024-2025	Retired from ChatGPT picker
GPT-5 (original)	Aug 2025	Retired
GPT-5.5 (Instant/Thinking/Pro)	2026	Active — current lineup

HuggingFace: Artificial Analysis LLM Leaderboard https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard



Each grey dot represents one model

If 1000 Wh = 20 hrs,
519 Wh = 10.38 hrs
If 1000 times = 10.38 hrs, 1 time = 0.01 hours or 37 secs
288 Wh = 21 secs

This is an exponential scale

New models will have different consumption

Sophia Chen. How much energy will AI really consume? The good, the bad and the unknown, Nature, 05 March 2025, https://www.nature.com/articles/d41586-025-00616-z. Summarisation = 0.5s, questions = 1.7s



It’s not a robot

The ‘elephants’ speak a language of probability

What you write:

“Who is the CEO of Apple?”

“Give me ideas for research about surveillance.”

What an LLM* does

***Predicts** a sequence of words that correlate with tokens **in its training data** clustered around ‘CEO’, ‘Apple’ or ‘research’, ‘idea’ and ‘surveillance” — and that a human is likely to **rate as useful**.*

*“Generative” is misleading: AI is “**translatative**”, mapping concepts in your prompt to concepts in its training.*

**LLM = large language model, e.g. GPT-5, Gemini 3.6 Flash, Sonnet 5*

It needs clarity; it uses harnesses

SN	Prompts	ChatGPT 4o	Gemini 2.5 Pro	Perplexity Pro
1	Guess a number between 1 - 50	27	37	27
2	Explain why you guess that number. Include references.	Provided explanation with references	Provided explanation with references	Provided explanation with references
3a	Generate 5 random numbers between 1 - 50. Explain the generated numbers, with references. Identify any relationship among the numbers. Verify that the references exist. Redo of inaccurate.	41, 8, 2, 48, 18	7, 13, 21, 34, 42	40, 29, 10, 25, 30
3b	Using Mersenne Twister, generate 5 random numbers between 1 - 50. Explain the generated numbers, with references. Identify any relationship among the numbers. Verify that the references exist. Redo	47, 27, 9, 38, 48	16, 49, 3, 22, 38	43, 1, 16, 28, 42
4	Guess 3 numbers between 1 - 50. Explain your guesses, with	17, 23, 7	42, 23, 37	6, 26, 4

“Guess vs Generate”

Source

The elephant has “jagged intelligence”

“AI tools demonstrate **excellent abilities on certain problems but surprising failures on other similar problems**. For humans, one kind of skill can often predict abilities at similar skills; this is not the case in the jagged landscape of AI.”

Melanie Mitchell. Jagged Intelligence: The dangerous unknowns at the heart of LLMs, The Yale Review

The epistemology of elephants (aka AI)

- Predictive nature of AI generates **most common statements**, not the most accurate or significant (bias towards canonical sources), e.g. Churchill
- Whose voices are most ‘common’ in the training data?
- Temporal, linguistic-geographical and sycophancy biases in training
- Responses and citations come in that order (not find source first)
- Inconsistent support from citations: “50% and 90% of LLM responses are not fully supported, and sometimes contradicted, by the sources they cite.”:
- Citation **quality** highly variable, e.g. YouTube, AI grooming, GEO
- Citation is “highly volatile”: Reddit went from 60% to 10%
- Significant % of sources cited by ChatGPT are **LLM-generated** (slop)
- Evidence that AI is **biased** towards LLM-generated material



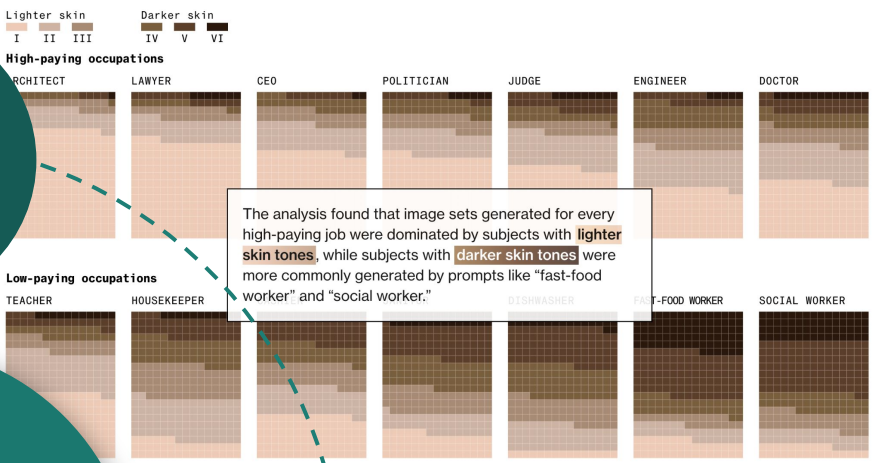
It’s not “biased”

‘Biases’

Biases in training data

Biases in design

User (your) biases

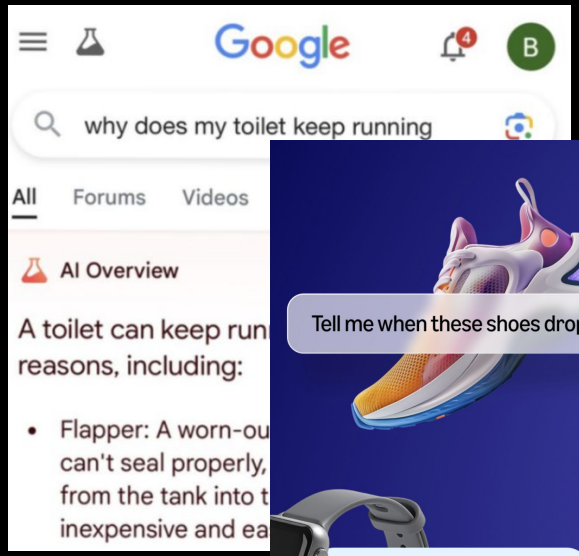


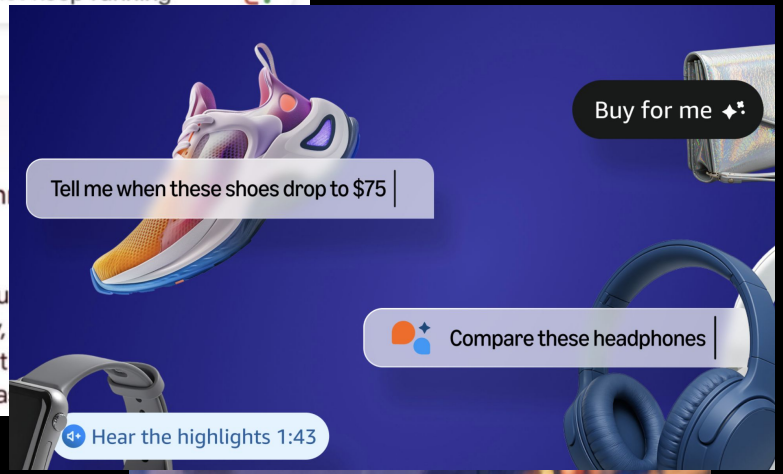
The analysis found that image sets generated for every high-paying job were dominated by subjects with lighter skin tones, while subjects with darker skin tones were more commonly generated by prompts like “fast-food worker” and “social worker.”

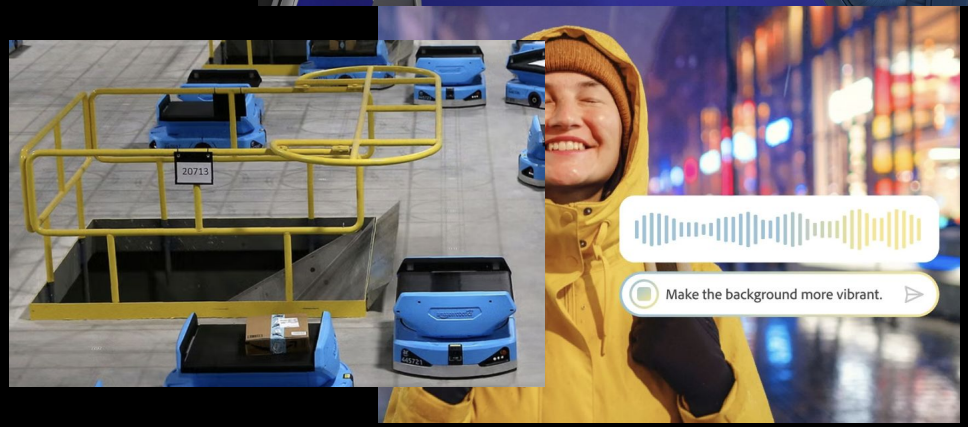




[Vox](#) / [Bloomberg](#) / [ChatGPT](#)







It's not a choice

The 12 fastest-growing skills in the UK

1. Relationship Building
2. **Strategic Thinking**
3. **AI Literacy**
4. Communication
5. **Large Language Model (LLMs) Utilisation**
6. Customer Relationship Management
7. **Adaptability**
8. Market Analysis
9. **Data Management**
10. Conflict Resolution
11. Documentation
12. **Regulatory Compliance**

LinkedIn Skills on the Rise 2025

Problem 1: We need to break down the challenges

1. **Identifying** AI usage
2. **Teaching** AI (critical issues, best practice, risk assessment)
3. **Assessing** AI (separating the machine from the human)
4. **Using** AI for research, where appropriate
5. **Risk** assessment (ethics, power, cost, deskilling, cognitive surrender)

Glossary

- **Accountability sink**: a system where no individual holds accountability, but instead a policy is held responsible
- **Automation bias**: “an overreliance on automated systems, favoring automated recommendations over their own judgment, even when contradictory and more accurate information is available” (Romeo & Conti)
- **Cognitive offloading**: moving cognitive work from an internal process to an external one. Examples include counting on your fingers (instead of your head), or storing a number in your phone memory (instead of memorising it). Cognitive offloading is not inherently bad: writing something down is cognitive offloading, and allows us to focus mental energy elsewhere. It’s a choice of where to focus mental effort.
- **Cognitive surrender**: “a deeper abdication of critical evaluation, where the user relinquishes cognitive control and adopts the AI's judgment as their own.” (Shaw & Nave)
- **Critahype**: hype that appears negative about AI but serves to inflate share price: for example hype about AI destroying millions of jobs serves to suggest larger growth potential for tech companies than predictions of augmenting existing roles.
- **Human in the loop (HITL)**: A human whose role is to check one or more steps in an automated process. *See also: moral crumple zones.*